

Topics

Implementation of Constraints Satisfaction Problem using suitable examples.

List of Problems

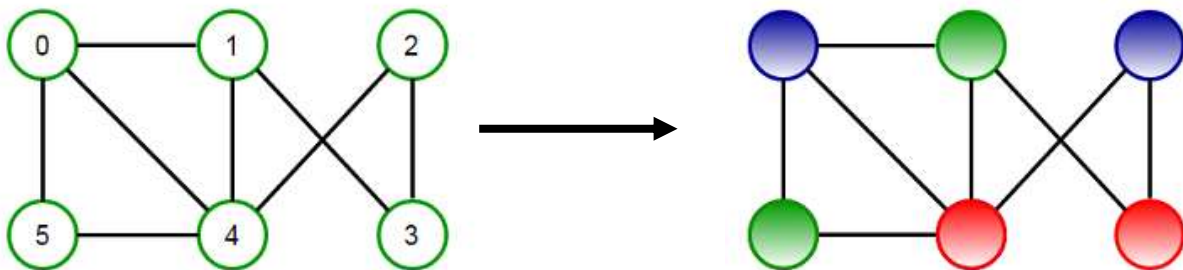
Problem 1. (Map Coloring)

Solve a map coloring problem using three colors ['blue', 'red', 'green']

Find all the combinations of assigning one of these colors to each territory.

Constraint: No territory can have the same color as any of its neighbors.

Constraint Graph:



Problem 2. (N-Queens)

Apply CSP Techniques (Filtering, Ordering, And Backtracking) to the 5-Queens Problem. Place 5 chess queens on a 5×5 chessboard so that no two queens attack each other.

		Q		
Q				
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Problem 3. (Cryptarithmic)

A Cryptarithmic problem,

Add both terms S E N D and M O R E to bring M O N E Y as a result.

$$\begin{array}{r} \text{S E N D} \\ + \text{M O R E} \\ \hline \text{M O N E Y} \\ \hline \end{array}$$

The rules or constraints on a Cryptarithmic problem are:

- There should be a unique digit to be replaced with a unique alphabet.
- The result should satisfy the predefined arithmetic rules, i.e., $2+2=4$, nothing else.
- Digits should be from 0-9 only.
- There should be only one carry forward, while performing the addition operation on a problem.
- The problem can be solved on both sides, i.e., left hand side (L.H.S), or right-hand side (R.H.S)